Introductions

DAEN 640 Capstone Senior Design

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Plan for today

Introductions

Goals & Expectations

Semester Overview

Grading & Assessment

Code of Conduct

Instructors

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Goals and Expectations

What is The Most Important Thing?

In life, the most important thing is subjective

Health, Love & Relationships, Purpose & Meaning, Gratitude, Freedom, Learning & Growth, Self-Care

These dimensions makes a conceptual backbone for our capstone, framing it not as a purely technical culmination experience, but as a professional & personal transition course

The Bridge Between Academic Training & Life (After Graduation)

Comprehensive technical expertise!

Learn to go beyond working your team's project, develop the ability to work in any project!

Success is fully achieved when combined with strong communication!


COMMUNICATION
THE KEY IS

Capstone Design represents the culmination of an educational journey

It's where **theory** meets **practice**, **abstraction** meets **reality**

The individual student transitions from learning concepts to creating, evaluating, and owning a complete system

Goals

Apply data engineering knowledge

Develop end-to-end project skills

Plan, document, execute projects using a structured framework (V-Model)

Foster collaboration & professional skills

Build a portfolio demonstrating technical & teamwork skills

Prepare for industry & research

Strengthen problem-solving & critical thinking

Experience working under professional expectations

Expectations

Active Participation

Ownership & Accountability

Professional Communication

Quality & Documentation

Ethical Conduct

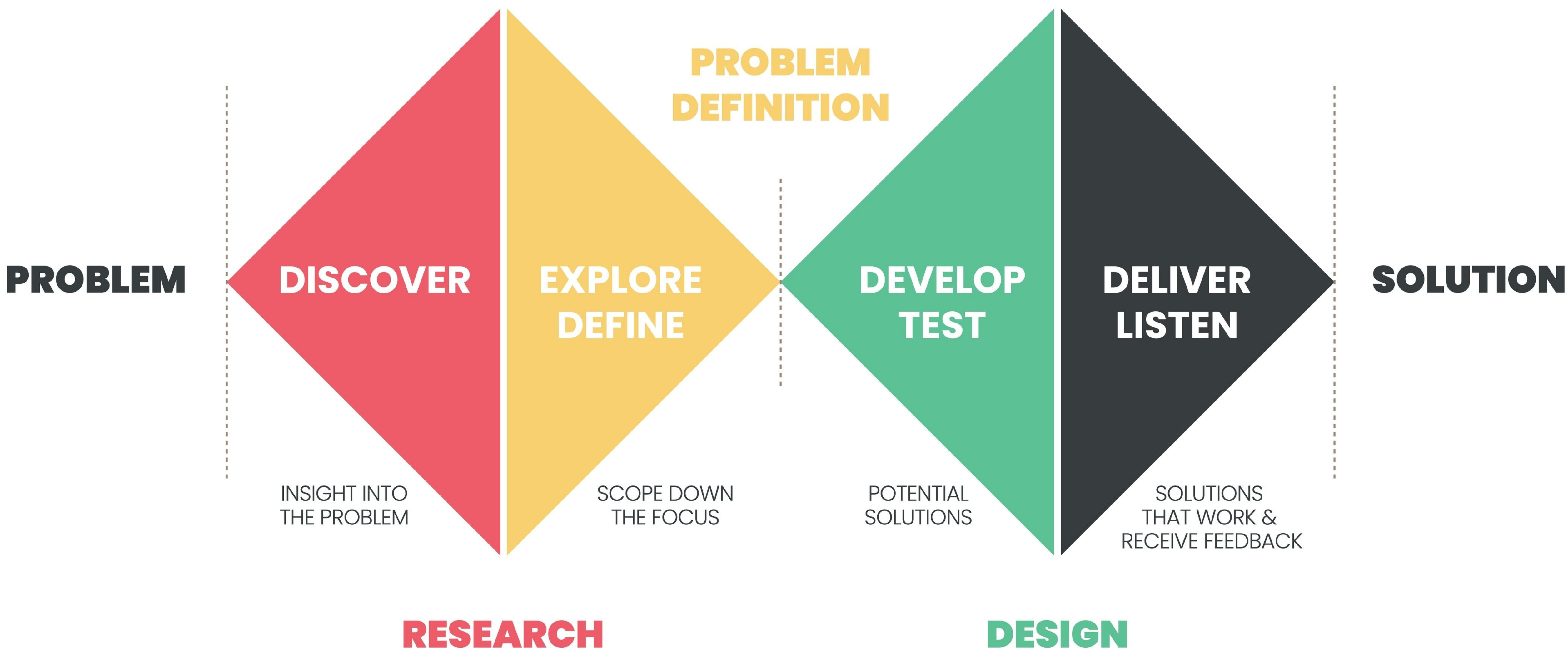
ISEN Mission Statement

> **To serve** the state, nation and global community by educating industrial engineering students to be well founded in engineering fundamentals and **to have the knowledge and skills** required **to design, develop, improve, implement and control** sophisticated production and service **systems** in an environment characterized by **complex technical and social challenges** . Throughout this educational process, students will be instilled with the **highest standards of professional and ethical behavior**.

INDUSTRIAL & SYSTEMS ENGINEERING
TEXAS A&M UNIVERSITY

Semester Overview

DESIGN THINKING
PROCESS



Discover: Understand the Problem

Purpose: Explore and learn before building

- Investigate the problem domain
- Identify stakeholders and users
- Study existing solutions and technologies
- Gather data (surveys, literature, benchmarks)
- Uncover unmet needs and constraints

Outcome: Clear understanding of what problem matters and why

Define: Frame the Right Problem

Purpose: Turn research into a focused project

- Synthesize findings into insights
- Write a precise problem statement
- Define scope, objectives, and constraints
- Establish functional and performance requirements
- Set success metrics and milestones

Outcome: A well-scoped, testable project definition

Develop: Design and Build

Purpose: Create and validate the solution

- Generate and compare design concepts
- Select the optimal solution
- Create system architecture and models
- Build prototypes or minimum viable products
- Test, iterate, and refine based on data

Outcome: A functional, validated engineering solution

Deliver: Validate and Communicate

Purpose: Prepare for real-world use and evaluation

- Perform final verification and validation
- Confirm performance meets requirements
- Finalize documentation, code, and design artifacts
- Prepare reports, posters, and presentations
- Demonstrate the solution to stakeholders

Outcome: A complete, defensible project ready for deployment and assessment

Weekly Schedule

Week	Date	Topic
1	1/13	Introductions + Comm Requirements
2	1/20	Integrate Engineering Design (*The Toaster Project*)

Weekly Schedule

Week	Date	Topic
3	1/27	Phase 1 Kickoff
4	2/3	Phase 1 Presentations - Groups BH & COE
5	2/10	Phase 1 Presentations - Groups ML
6	2/17	Phase 2 Kickoff
7	2/24	Phase 2 Presentations - Groups BH & COE
8	3/3	Phase 2 Presentations - Groups ML

March 7-15

Weekly Schedule

Week	Date	Topic
9	3/17	Phase 3 Kickoff
10	3/24	Phase 3 Presentations - Groups BH & COE
11	3/31	Phase 3 Presentations - Groups ML
12	4/7	Phase 4 Kickoff
13	4/14	Phase 4 Presentations - Groups BH & COE
14	4/21	Phase 4 Presentations - Groups ML

Attendance

- Mandatory attendance for everyone:
  - First two weeks of class
  - Phase kick-offs
- Team presentation days:
  - All team members must stay for the entire class!
  - Other students are welcome & encouraged to attend

Again

Phase Kickoff Weeks

- Entire team must be present
- Details on phase
  - Content
  - Deliverables (checklist)
- Communications Activities
- Project guidance/tips

And Again

Presentation Weeks

- On the day for your group, ALL team members must be present, the entire time!
  - Appropriate attire
  - Be a good listener
  - Ask questions, participate!
- You are welcomed/encouraged to attend presentations on the day of the other group

Faculty Advisor

Each team must select a Texas A&M faculty advisor.

- Advisors are chosen for their expertise or interest in the project topic.
- Instructors may assist in identifying advisors if needed.

Faculty Advisor

- Advisors serve as the team's subject matter expert, providing guidance & recommendations.
- Biweekly meetings are recommended, based on advisor availability.
- Advisors take an active advisory role but are not responsible for developing solutions.
- Teams must treat advisors with professional respect & courtesy.

Sponsors

- The project sponsor is the **client**, owner of the project problem or opportunity.
- Each team must assign a *Point of Contact* to manage sponsor communication.
- Communication frequency & format should be defined based on sponsor availability.
- Regular **face-to-face** meetings (in person or video) are required; email alone is insufficient!

Sponsors

ALL team members should attend sponsor meetings; absences must be acknowledged with reasons

- Meetings are scheduled by the POC and sponsor and must start and end on time with an agenda.
- Only the sponsor may cancel meetings; cancellations must be reported in technical presentations.
- If meeting frequency is unacceptable to the sponsor, document it and consult the primary instructor.

Confidentiality & Non-attribution

- Course data, client info, and project discussions are confidential
  - Share information only with classmates or instructors, following confidentiality rules
- Keep confidential materials secure
  - In a folder, binder, or password-protected device
  - On your person or in a safe location (room, backpack, car)

Confidentiality & Non-attribution

- Don't attribute information to specific people or teams
  - Use phrases like: "It's my understanding that..."
- Treat NDAs seriously!

Grading & Assessment

Team Grades	Weight on the Final Grade	Grade Type
Mini Management Packet	5%	Technical
Phase 1-3 Presentations	15% (5% each)	"
Phase 4 Presentation	10%	"
Showcase Poster	5%	"
Communication Log	2% (1% each)	"
Final Report	18%	"
Individual Grades		
Phase Presentations	15%	Communication
Writing Assignments	18%	Writing
Peer Performance	12%	

Deliverables

- Code of Conduct: will not be graded but MUST be turned in as a group assignment!
  - As well other things in the deliverables table: Showcase Registration, Meeting Logs, etc.
- Next week: more details on the Engineering Showcase, Phase Presentations, etc.

Deliverables

- Check Canvas for all due dates
  - They're final unless we send a notice by Canvas + email
- You can resubmit assignments for a better grade after the due date
- All regrade work must be in by the assignment's close date
- When resubmitting, make sure to:
  - Highlight what you changed
  - Address ALL feedback from class, presentations, or office hours

Use AI responsibly and within limits

- Brainstorm ideas
- Check your work for grammar, spelling, clarity, and brevity

AI = tool, not shortcut!

AI should support learning, not replace it!
Always ask questions and participate in class

Use AI in ways you could confidently explain to your instructor or classmates.

You may NOT use AI to

- Write assignments for you
- Solve major problems or programming tasks
- Generate analyses or substantive content

All work must reflect your own effort
Misuse may be treated as academic dishonesty!

Do NOT input confidential sponsor information into AI tools

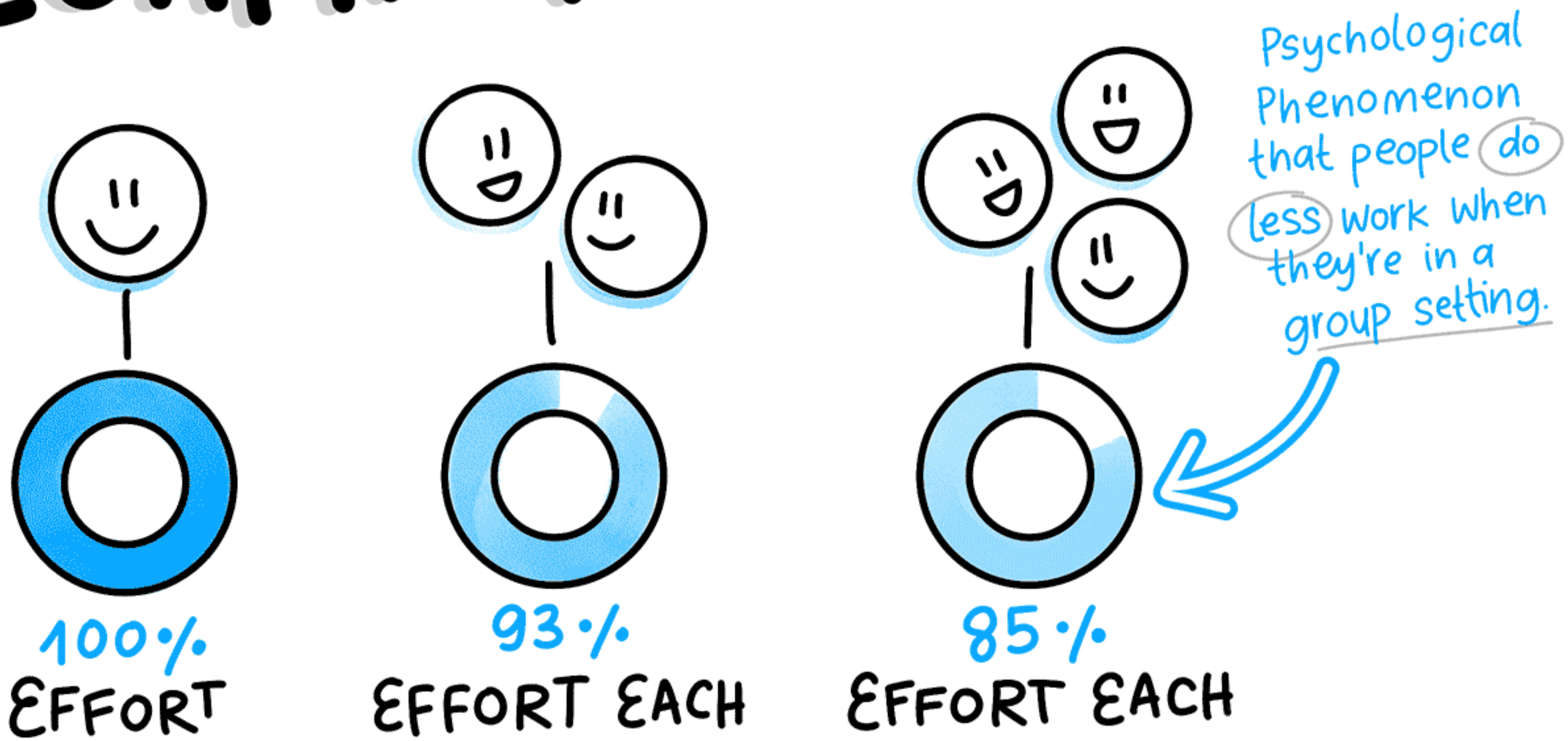
Team Work

In industry, project failure is rarely caused by lack of technical skill.

Most often caused by:

- Unclear expectations
- Poor communication
- Uneven workload distribution
- Missed deadlines
- Unresolved conflict
- Unprofessional behavior

What is SOCIAL LOAFING? — How does it impact teams?

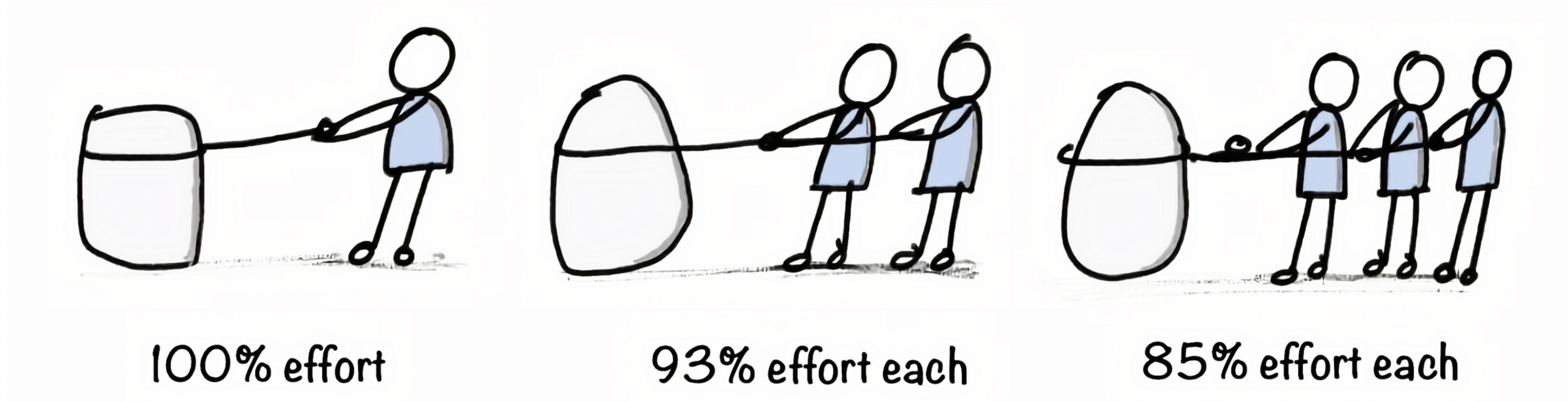


Examples of social loafing

- Disappearing after class and not responding to messages.
- Continuing poor communication even after being made aware of the issue.
- Expecting the team to adjust around your personal schedule.
- Missing meetings, arriving late, or leaving early on a regular basis.
- Failing to complete assigned work, or turning in work that is incomplete, careless, or must be redone by teammates.
- Working on off-topic ideas without team approval and expecting credit.
- Attending meetings but being distracted (by phone instead of participating).

All team members MUST be reliable, communicative, prepared, and engaged!

<u>Social Loafing</u> means not contributing your fair share to team work



This behavior creates serious problems for teams and is taken very seriously in this course

Code of Conduct

Code of Conduct

Let us prevent these problems BEFORE they occur

On Canvas - Due 1/19 at 5 PM

https://canvas.tamu.edu/courses/431435/assignments/2874310

Professional Standards of Conduct

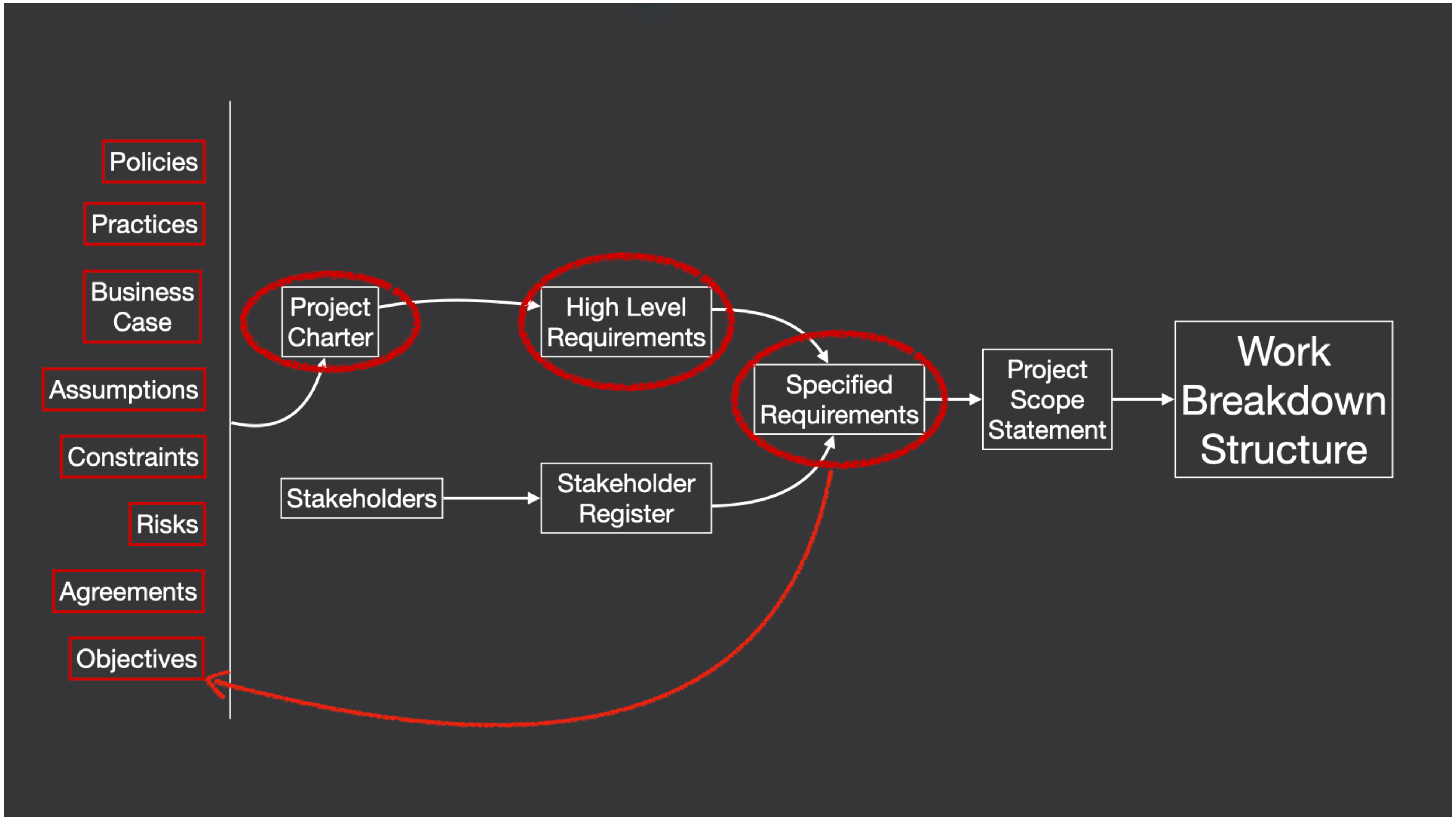
- Learning includes respectfully engaging with different opinions & keeping an open mind.
- All in this course is expected to treat others with **courtesy, respect, and professionalism**.
- These expectations apply to **all** course-related activities & meetings.
- Communication should be factual, constructive, and free of harassing, offensive, or demeaning language.

Professional Standards of Conduct

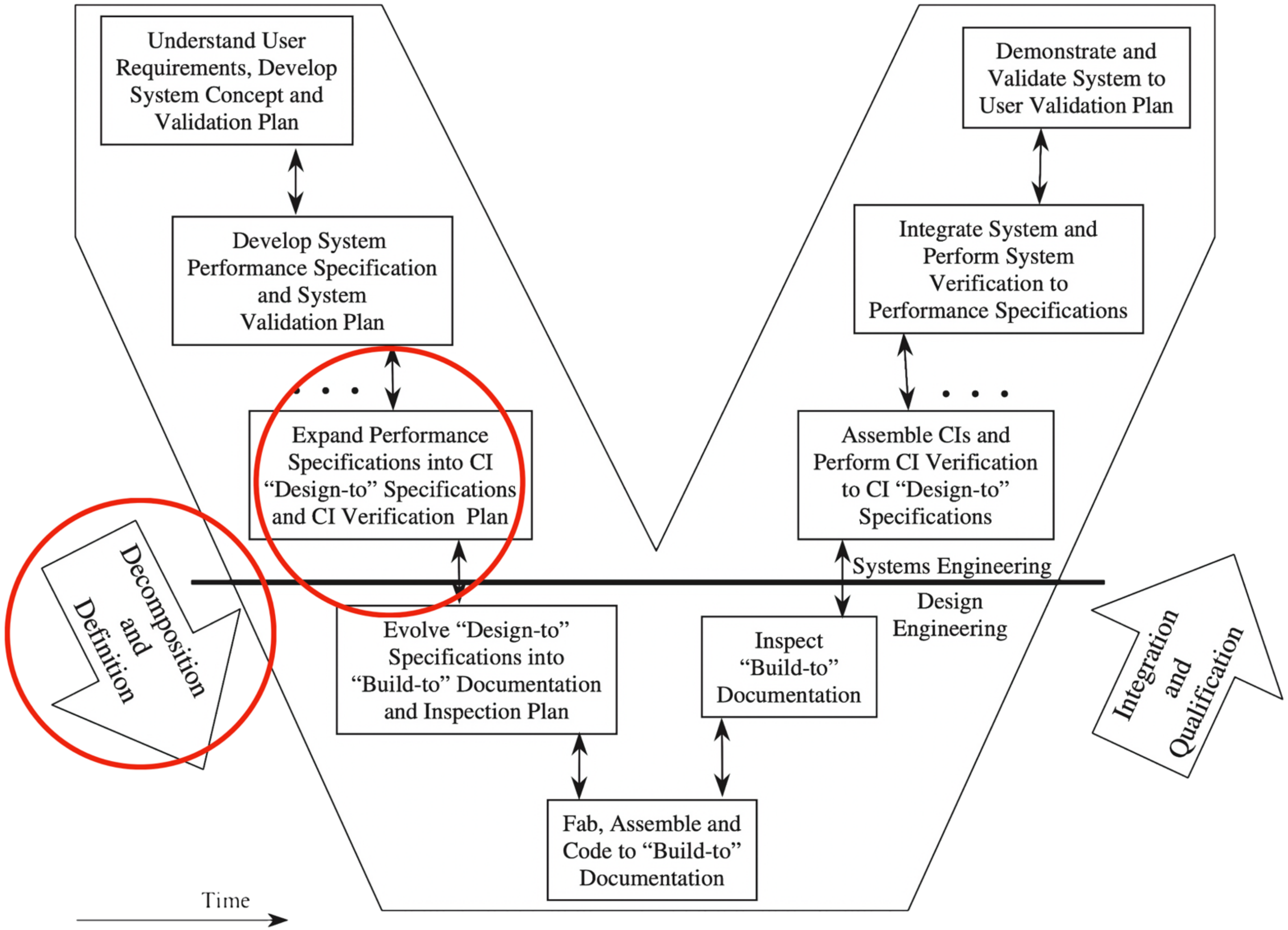
- Disagreements are allowed,
  ... but they must be fact-based & civil in tone.
- Abusive, disruptive, threatening, harassing behavior and profanity are **NOT** acceptable.
- If an issue occurs, individuals will usually be informed and guided on appropriate behavior, and are expected to correct it.
- Serious violations may result in University disciplinary action.
- Students are responsible for knowing and following all student rules and policies.

Next Week Preview

Project Charter Adjustments



V-Model



Understand User Requirements, Develop System Concept and Validation Plan
Develop System Performance Specification and System Validation Plan
Expand Performance Specifications into CI "Design-to" Specifications and CI Verification Plan
Evolve "Design-to" Specifications into "Build-to" Documentation and Inspection Plan
Fab, Assemble and Code to "Build-to" Documentation
Inspect "Build-to" Documentation
Assemble CIs and Perform CI Verification to CI "Design-to" Specifications
Integrate System and Perform System Verification to Performance Specifications
Demonstrate and Validate System to User Validation Plan
Systems Engineering
Design Engineering
Decomposition and Definition
Integration and Qualification
Time

What is a Configuration Item?

What is a Configuration Item in Data Engineering?

What is a Configuration Item (CI)?

Any hardware, software, firmware, document, or combination thereof that is:

Individually identifiable

Formally baselined

Controlled under configuration management

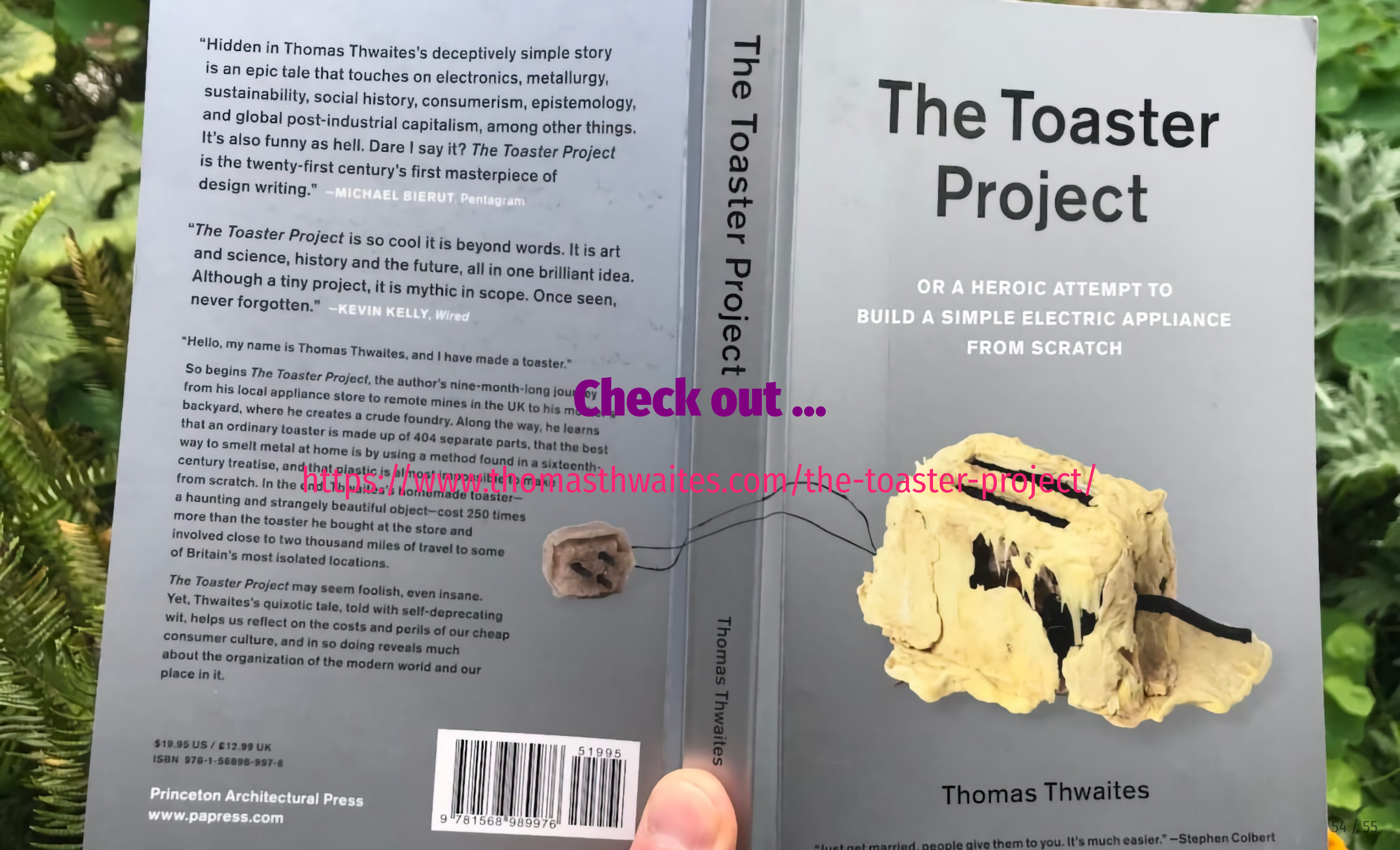
Verified against approved specifications

In other words, a CI is anything the organization has decided must be controlled, versioned, and verified as its own official product element.

What Counts as a CI in a Data Platform

- In data engineering, a CI is:
  - Any version-controlled, independently testable, production-quality component of a data system that has formal requirements, formal tests, and formal approval.
- A CI is no longer "just a file", it becomes a certified system component.

CI Area	Examples
Sources	Ingest schema, API feed
Pipelines	ETL job, stream proc
Storage	Warehouse schema, lakehouse
Code	Feature lib, validation
Models	Forecast, classifier
Orchestration	Airflow DAG, dbt
Interfaces	API, data contracts
Governance	Quality rules, PII mask
Docs	Data dict, lineage


Check out ...
https://www.thomasthwaites.com/the-toaster-project/
The Toaster Project
OR A HEROIC ATTEMPT TO
BUILD A SIMPLE ELECTRIC APPLIANCE
FROM SCRATCH
Thomas Thwaites
"Hidden in Thomas Thwaites's deceptively simple story is an epic tale that touches on electronics, metallurgy, sustainability, social history, consumerism, epistemology, and global post-industrial capitalism, among other things. It's also funny as hell. Dare I say it? The Toaster Project is the twenty-first century's first masterpiece of design writing." —MICHAEL BIERUT, Pentagram
"The Toaster Project is so cool it is beyond words. It is art and science, history and the future, all in one brilliant idea. Although a tiny project, it is mythic in scope. Once seen, never forgotten." —KEVIN KELLY, Wired
"Hello, my name is Thomas Thwaites, and I have made a toaster."
The Toaster Project may seem foolish, even insane. Yet, Thwaites's quixotic tale, told with self-deprecating wit, helps us reflect on the costs and perils of our cheap consumer culture, and in so doing reveals much about the organization of the modern world and our place in it.
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"Just get married, people give them to you. It's much easier." —Stephen Colbert

Make it a GREAT semester! ツ